

03_Loops_in_Python

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0.0.1 For Loop

```
[1]: N=10 #number of iterations  
  
for i in range(N):  
    print(i)
```

0
1
2
3
4
5
6
7
8
9

```
[2]: for i in range(5,10):  
    print(i)
```

5
6
7
8
9

```
[4]: #how to print numbers from 5 to 10  
for i in range(5,10+1):  
    print(i)
```

5
6
7
8
9
10

```
[5]: #several lines inside a block
cumulative=0
for i in range(1,10):
    x=i**2
    cumulative=cumulative+x
print('result is',cumulative)
```

result is 285

The second option is when i have an exiting list (or other iterable) with elements. The loop will go through the elments of this list.

```
[7]: mylist=['car', 'train', 'bike', 'boat' ]

for t in mylist:
    print(t)
```

car
train
bike
boat

```
[8]: #print how many letters has each word

for w in mylist:  # i use w now, as in 'word'
    len_w=len(w)
    print('the word',w,'has',len_w,'letters')
```

the word car has 3 letters
the word train has 5 letters
the word bike has 4 letters
the word boat has 4 letters

```
[9]: # iterating over a tuple

mytup=(1,6,5,9)
for n in mytup:
    print(n)
```

1
6
5
9

```
[10]: # iterating over a string

mytext='This is a text'
for c in mytext:  # 'c' as in 'character'
    print(c)
```

T
h
i
s

i
s

a

t
e
x
t

```
[11]: # we can also iterate over a sequence
      # a known sequentially at which iteration we are, using enumerate

      namelist=['Peter', 'Maria', 'Emily', 'Steven' ]

      for i,name in enumerate(namelist): # look that returns two things
          print(i, name)
```

0 Peter
1 Maria
2 Emily
3 Steven

A more interesting example. iterating over rows of a pandas dataframe

```
[3]: import pandas as pd
      df= pd.read_excel('./data/CashFlow.xlsx', sheet_name='CF_Scenario2')
      df.head()
```

```
[3]:
```

	Year	CF	discount rate
0	0	-1000000	0.050
1	1	200000	0.040
2	2	200000	0.040
3	3	200000	0.043
4	4	200000	0.045

```
[5]: from math import exp
      CF=2000
      rate=.04
      T=2
      PV=CF*exp(-rate*T)
      PV
```

```
[5]: 1846.2326927732715
```

```
[11]: for i, row in df.iterrows():  
       print(row, '\n')
```

```
Year          0.00  
CF          -1000000.00  
discount rate    0.05  
Name: 0, dtype: float64
```

```
Year          1.00  
CF           200000.00  
discount rate    0.04  
Name: 1, dtype: float64
```

```
Year          2.00  
CF           200000.00  
discount rate    0.04  
Name: 2, dtype: float64
```

```
Year          3.000  
CF           200000.000  
discount rate    0.043  
Name: 3, dtype: float64
```

```
Year          4.000  
CF           200000.000  
discount rate    0.045  
Name: 4, dtype: float64
```

```
Year          5.000  
CF          -500000.000  
discount rate    0.046  
Name: 5, dtype: float64
```

```
Year          6.000  
CF           200000.000  
discount rate    0.046  
Name: 6, dtype: float64
```

```
Year          7.000  
CF           200000.000  
discount rate    0.046  
Name: 7, dtype: float64
```

```
Year          8.000  
CF           200000.000  
discount rate    0.046  
Name: 8, dtype: float64
```

Year 9.000
CF 20000.000
discount rate 0.046
Name: 9, dtype: float64

Year 10.000
CF -500000.000
discount rate 0.046
Name: 10, dtype: float64

Year 11.000
CF 200000.000
discount rate 0.047
Name: 11, dtype: float64

Year 12.0000
CF 200000.0000
discount rate 0.0478
Name: 12, dtype: float64

Year 13.0000
CF 200000.0000
discount rate 0.0478
Name: 13, dtype: float64

Year 14.0000
CF 200000.0000
discount rate 0.0478
Name: 14, dtype: float64

Year 15.0000
CF -500000.0000
discount rate 0.0478
Name: 15, dtype: float64

Year 16.0000
CF 200000.0000
discount rate 0.0478
Name: 16, dtype: float64

Year 17.0000
CF 200000.0000
discount rate 0.0478
Name: 17, dtype: float64

Year 18.00
CF 200000.00
discount rate 0.05

Name: 18, dtype: float64

Year 19.00
CF 200000.00
discount rate 0.05
Name: 19, dtype: float64

Year 20.00
CF -500000.00
discount rate 0.05
Name: 20, dtype: float64

Year 21.00
CF 200000.00
discount rate 0.05
Name: 21, dtype: float64

Year 22.00
CF 200000.00
discount rate 0.05
Name: 22, dtype: float64

Year 23.00
CF 200000.00
discount rate 0.05
Name: 23, dtype: float64

Year 24.00
CF 200000.00
discount rate 0.05
Name: 24, dtype: float64

Year 25.00
CF 0.00
discount rate 0.05
Name: 25, dtype: float64

```
[12]: NPV=0
      for i, row in df.iterrows():
          CF=row['CF']
          rate=row['discount rate']
          T=row['Year']
          this_PV=CF*exp(-rate*T)
          NPV=NPV+this_PV
      print(NPV)
```

```
print('final NPV:',NPV)
```

```
-1000000.0  
-807842.1121695354  
-623218.8428922083  
-447424.04978309164  
-280370.0075008372  
-677636.8087525042  
-525874.2226002559  
-380934.5845342753  
-242511.14819652922  
-229291.12917121098  
-544932.951924674  
-425671.5943978898  
-312972.95293159166  
-205534.58446726028  
-103110.96277942206  
-347218.31543193816  
-254132.91401837667  
-165392.3265046298  
-84078.39455650999  
-6730.189865609747  
-190669.9104513309  
-120682.36062909984  
-54108.14388948394  
9219.209986326692  
69458.0523687671  
69458.0523687671  
final NPV: 69458.0523687671
```

```
[ ]:
```